

Hybrid IWD-DE: A Novel Approach to Model Cooperative Navigation Planning for Multi-robot in Unknown Dynamic Environment

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Abstract

The current study emphasizes inadequacies of earlier research findings on multi-robot navigation and offers a robust approach to guide the robots in an unknown dynamic workspace. To attain the objective, the Intelligent Water Drop (IWD) algorithm is combined with Differential Evolution (DE) in a new conceptual fashion to optimize the navigation path of multiple mobile robots. The hybrid IWD-DE, basic IWD and DE algorithms are then explored for multi-robot navigation in a dynamic workspace through simulation. The simulation outcomes demonstrate the competency of the proposed method to guide the robot in predicting the nature of the obstacle and to generate an optimal and safe trajectory in a dynamic workspace. The potency of the proposed algorithm is further validated through experimentation with fire bird robot in a real platform. The outcomes of the simulation and experimentation, by employing IWD-DE algorithm are compared with those of basic IWD and DE algorithm. The error estimated for Average Navigation Path Travelled (ANPT), Aggregate Navigation Path Deviation (ANPD) and number of turns between simulation and experimental results by employing IWD-DE is 8.96, 9.09, and 11.13, respectively while reaching 16.43, 28.61, 29.15 for IWD, and 6.66, 10.20, 14.28 for DE. The efficiency of the proposed approach is further verified with the state-of-the-art in a dynamic workspace. The outcomes reveal the efficiency and flexibility of the proposed approach in resolving the multi-robot navigation problem as compared to the state-of-the-art in a dynamic workspace.

Keywords: multi-robot, dynamic environment, intelligent water drop, differential evolution, navigation planning, waypoints, fire bird robot

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1 Introduction

Over the past few decades, mobile robotics emerged as the most trending and cost effective application in many fields such as industrial production^[1], logistics^[2], space research^[3], underwater vehicle^[4], surveillance^[5], agriculture^[6], search and rescue operation^[7] and many more. The main concern in the field of mobile robotics is to find safe and optimal navigation path for robots. Many research contributions have been made in this direction to boost the efficiency of mobile robot, such that it can arrive its target with minimum path^[8], time^[9] and energy usages^[10]. The workspace for robotic applications may be characterized as static or dynamic. In static, the position of target and obstacles are fixed, whereas, the position of target or obstacles may change in dynamic. Furthermore, based on the robot's knowledge about the workspace, its navigation planning may be categorized as online or offline. In

offline planning, the robot has the complete knowledge about the workspace and plans a safe path before starting its journey. Conversely, in online planning, the robot has partial knowledge about the workspace and finds the path on the go. Offline planning is more suitable for static workspace. However, combination of both approaches^[11] may be employed to enhance the performance of robot in finding an optimal and safe path in dynamic workspace. The research works contributed in these directions are outlined as follows.

The methodologies for solving mobile robot navigation problem may be termed as either classical or heuristics. The classical approaches employed for mobile robot navigation includes roadmap^[12–15], cell decomposition^[16] and potential field^[17,18]. However, these approaches are not efficient in the presence of dynamic obstacles, multi-robot environment and large obstacles. Because of these limitations, researchers put emphasis on heuristic approaches such as A-star, Simulated an-

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nealing, Neural Networks (NN), Fuzzy Logic (FL), population based algorithms and hybrid algorithms.

The issues related to navigation of mobile robot using A-star algorithm and its improvement has been addressed in Refs. [19–21]. However, the proposed approaches are not effective for online planning in dynamic workspace. Trajectory planning of mobile robot in dynamic workspace using Simulated Annealing (SA) is addressed by Refs. [22, 23]. SA algorithm is inspired by the annealing process of solid metal, where a smelted is cooled gradually to reduce defects. Apart from SA, FL is widely employed in optimal trajectory finding of mobile robots which mimics the behavior or perception of human being to deal with the uncertainties in the environment. Mobile robot navigation strategies employing Fuzzy Logic Controller (FLC)^[24–27] are critically examined in both static and dynamic workspaces. Despite the ability of FL, the selection of appropriate membership function and formation of fuzzy rules by FLC does not always confirm in obtaining the optimal path. The another extensively used approach in trajectory planning of mobile robot is Neural Network (NN)^[28–31], which uses the sensory information such as distance of obstacle from its front, left and right position as input and finds the steering angle as output. The performance of NN mostly depends on proper selection of number of hidden layers, number of neurons per each layer, and adjustment of weights which in effect affects the optimum solution. The computational error and execution time involved in above discussed approaches needs further refinement in results, which pave the way for employing meta-heuristic approaches in resolving the trajectory planning of mobile robot. Some notable approaches such as Genetic Algorithm (GA)^[32–34], Gravitational Search Algorithm (GSA)^[35,36], Particle Swarm Optimization (PSO)^[37–41], Ant Colony Optimization (ACO)^[42] and Bat Algorithm (BA)^[43] are found to be reliable in addressing the issues related to robot navigation. However, the above cited approaches mostly contribute in addressing the single robot navigation. Although the discussed methods deliver a good estimation of the optimal solution, it is not efficient for multi-robot navigation planning in the presence of dynamic obstacles. A survey on Navigation planning of mobile robot in dynamic workspace is critically studied

in Ref. [44]. Moreover, the efficiency of the proposed methods has not been verified in real environment and it is not efficient in maintaining the balance between exploration and exploitation capability while optimizing the path length. The real scenario demands a number of tasks to be accomplished in a certain time period. Thus, multiple mobile robots are employed to carry out the tasks to overcome the inefficiency of single robot in a stipulated time period. However, finding safe and optimal navigation path for multiple mobile robots in an unknown dynamic environment are one of the challenging issue. Looking into the complexities involved in Multi-Robot System (MRS) it is indispensable to design a robust method to deal with these challenges.

Thus, a novel hybrid IWD-DE algorithm has been proposed to deal with multi-robot navigation problem in a dynamic workspace. The proposed algorithm utilizes the inherent behavior of water drops in the river bed to guide the robot in finding the shortest path. For better exploration capability of IWD, it is further tuned using DE operator. The proposed approach makes use of a novel methodology for anticipating the behavior of obstacles in the environment. A single objective fitness function comprising the weighted sum of path distance, path smoothness and path safety is formulated and evaluated consistently by the algorithm to generate an optimal and safe trajectory for mobile robots. To the best of authors' acquaintance, IWD has been used to address single robot navigation in a static environment. Thus, it motivated us to develop a hybrid IWD-DE algorithm with new features to address the multi-robot navigation problem. The remaining of this paper is structured as follows;

Section 2 outlines the problem formulation for navigation planning of multi-robot in a dynamic workspace. Section 3 exhibits the fundamental aspects regarding hybridization of IWD-DE algorithm and implementation of it in navigation planning of multiple mobile robots. The experimental setup and performance evaluation parameters for validating the proposed approach are discussed in section 4. The effectiveness of the proposed approach in comparison to the state-of-the-art is validated in section 5. Finally, the concluding remark with further scope is highlighted in section 6.

2 Problem formulation

The fitness function formulation for navigating the multiple mobile robots in a dynamic workspace with the prediction of obstacles behavior is analyzed in this section. The problem of interest is to find an optimal and safe path for two wheeled mobile robots in dynamic workspace consist of polygonal obstacles. The following accompanying presumptions must be made for the problem.

- (a) Every robot knows their respective initial and target position in the workspace.
- (b) The orientation of every robot is towards the respective target position before the start of navigation.
- (c) Every robot starts their navigation planning from their respective initial position at the same time.
- (d) The dynamic obstacles are moving in a particular direction with constant velocity and less than the velocity of robot.
- (e) All the mobile robots in the workspace are of identical structure.

The position of a mobile robot at particular instance of time in the 2-dimensional workspace can be defined using a triplet (x_c, y_c, θ) , where (x_c, y_c) is the position of robot with respect to XY -plane and θ is the orientation of the mobile robot. The navigation path of a mobile robot can be represented by series of waypoints as shown in Fig. 1. Let, “wp” be the number of waypoints in between the source (S_i) and target (G_i) position of i th robot. Thus, the total path traveled by the i th robot (P^i) is the sequence of sub paths and may be expressed as $P^i = \{p_0^i - p_1^i, p_1^i - p_2^i, \dots, p_{wp}^i - p_{wp+1}^i\}$, where, p_0^i and p_{wp+1}^i are the source and target position of i th robot respectively. If (x_j^i, y_j^i) is the coordinate of j th waypoint of i th robot then the navigation path traveled by the robot can then be expressed as:

$$f_{\text{path}}^i = \min \left\{ \sum_{j=0}^{wp} \sqrt{(x_{j+1}^i - x_j^i)^2 + (y_{j+1}^i - y_j^i)^2} \right\}. \quad (1)$$

Further, the smoothness of the path depends on the

waypoints *i.e.*, more the number of waypoints more is the smoothness of the path. On the contrary, the robot will consume more energy with increase in waypoints on its path. In consequence of this, the number of waypoints should be minimized to achieve maximum smoothness of the path. For example, assume there are two possible navigation paths $[p_0-p_1-p_3]$ and $[p_0-p_2-p_3]$ with one intermediate waypoint p_1 and p_2 respectively as shown in Fig. 2. Initially the robot’s heading direction is towards the goal. The maximum angle of deviation for robot to reach its target by choosing the path $[p_0-p_1-p_3]$ and $[p_0-p_2-p_3]$ is $\theta_1+\theta_2$ and $\theta_3+\theta_4$ respectively. However, the path with waypoint p_1 is smoother than the path with waypoint p_2 as $\theta_1+\theta_2 < \theta_3+\theta_4$. Thus, the path smoothness can be achieved by minimizing the number of waypoints while maximizing the angle of deviation and the objective function for it can be expressed as:

$$f_{\text{smooth_wp}}^i = \min \{wp_i\}, \quad (2)$$

$$f_{\text{smooth_ang}}^i = \min \left\{ \max_{0 \leq j \leq wp-1} \{Angle[p_j^i, p_{j+1}^i, p_{j+2}^i]\} \right\}, \quad (3)$$

where $Angle[p_j^i, p_{j+1}^i, p_{j+2}^i]$ is the angle of deviation between the sub paths $p_j^i - p_{j+1}^i$ and $p_{j+1}^i - p_{j+2}^i$ of the i th robot and can be computed as Eq. (4).

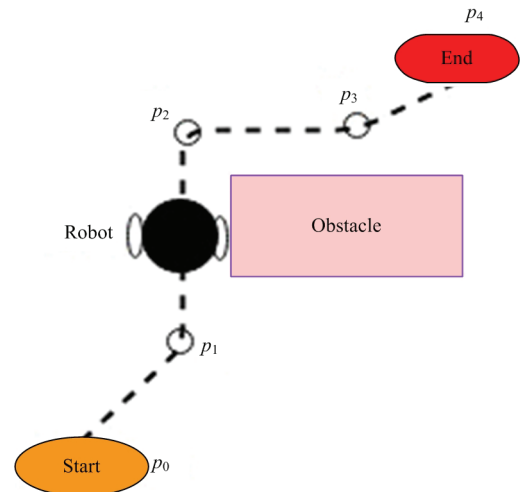


Fig. 1 Waypoints in the robot’s path.

$$Angle[p_j^i, p_{j+1}^i, p_{j+2}^i] = \pi - \cos^{-1} \left(\frac{(x_{j+1}^i - x_j^i)(x_{j+2}^i - x_{j+1}^i) + (y_{j+1}^i - y_j^i)(y_{j+2}^i - y_{j+1}^i)}{\sqrt{(x_{j+1}^i - x_j^i)^2 + (y_{j+1}^i - y_j^i)^2} \times \sqrt{(x_{j+2}^i - x_{j+1}^i)^2 + (y_{j+2}^i - y_{j+1}^i)^2}} \right). \quad (4)$$

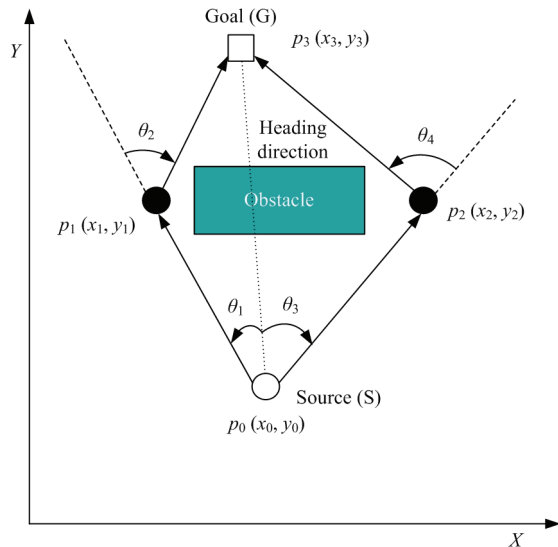


Fig. 2 Illustration for smoothness of path with different angle of deviation.

Besides the shortest and smooth path, path safety is utmost important in navigation planning of mobile robot. In order to achieve the path safety, a mobile robot should maintain a minimum distance from the obstacles. Such as to navigate safely in the unknown workspace, it is necessary for the robot to anticipate the behavior (static or dynamic) of obstacles.

The position and behavior of the obstacles in the workspace is unknown to the mobile robot; therefore, the periphery of the robot is uniformly installed with sensors to acquire information from its surrounding. To execute the task of predicting obstacle position and behavior, the sensor range of the mobile robot is divided into four circles with radii r_1 , r_2 , r_3 and r_4 as shown in Fig. 3^[45]. The radii r_4 and r_1 represent the maximum sensing distance and the safe distance to be maintained by the robot to avoid collision, respectively. Let an obstacle penetrate the sensing range of the robot at time t , while the position of the robot is (x_r^t, y_r^t) . The position of obstacle at any instant of time t can be obtained through distance of obstacle from the robot. In a similar manner, the position of obstacle with respect to robot at time $t + \Delta t$ can be obtained. If the predicted position of the obstacle with respect to XY -plane at time t and $t + \Delta t$ remain unchanged, then the obstacle will be treated as static otherwise it is dynamic. If the obstacle is dynamic, then the prediction of its velocity and direction is of paramount importance. The displacement of obstacle in

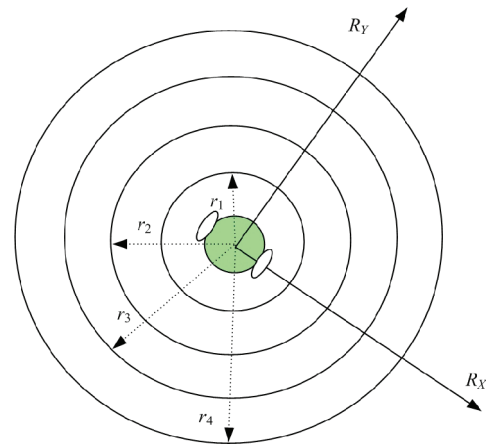
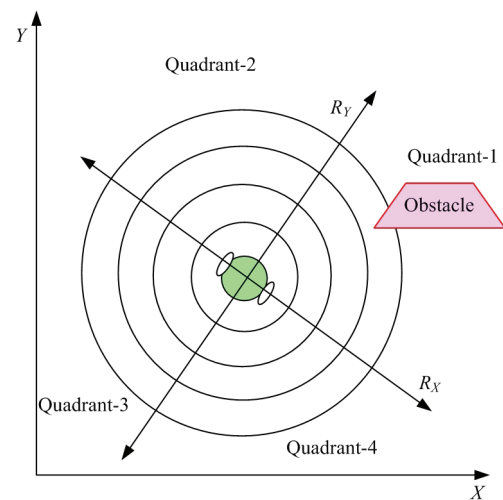
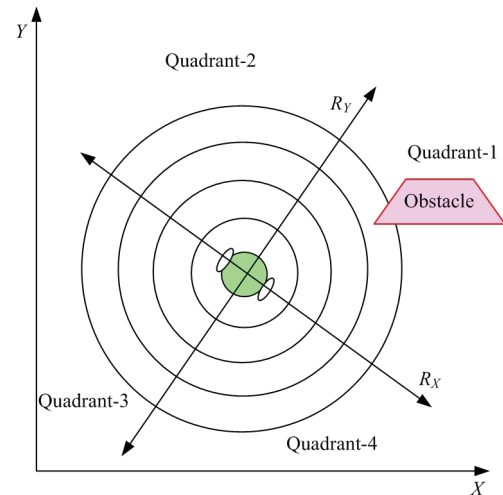


Fig. 3 Sensor range of mobile robot.



(a) At time t



(b) At time $t + \Delta t$

Fig. 4 Detection of obstacle behavior.

time Δt is the measure of obstacle's velocity (v_o). Further, the circular sensing area of the mobile robot is divided

into four quadrants as shown in Fig. 4 to obtain the direction of moving obstacle. The direction of motion of obstacle mainly depends on the number of intersection points. The obstacle moves towards the robot if the number of intersecting points increases otherwise it is away from the robot. The robot needs to decide the safe navigation path when a dynamic obstacle is moving towards the robot. The dynamic obstacle avoidance problem can be transformed into a static obstacle avoidance problem by using the concept of relative velocity. That is, the computation for obstacle avoidance between the mobile robot moving with a velocity $\mathbf{v}_r$ and obstacle moving with constant velocity $\mathbf{v}_o$ at time t is similar to computing the obstacle avoidance between the mobile robot moving at a velocity $\mathbf{v}_{ro}$ and a static obstacle. The scenario of relative velocity is shown in Fig. 5.

Let $\mathbf{v}_r$ and $\mathbf{v}_o$ are the velocity of the robot and the obstacle respectively. Thus, the relative velocity of the robot with respect to the obstacle ($\mathbf{v}_{ro} = \mathbf{v}_r - \mathbf{v}_o$) can be estimated using:

$$\mathbf{v}_{ro} = \frac{\min_{1 \leq i \leq k} \{dis_o^i(t + \Delta t) - dis_o^i(t)\}}{\Delta t}, \quad (5)$$

where $dis_o^i(t + \Delta t)$ and $dis_o^i(t)$ are the distance measured by the i th sensor ($i = 1, 2, \dots, k$) at time $t + \Delta t$ and t , respectively. The value of relative velocity gives additional information regarding the expected time of collision as per Eq. (6). It provides a mean to take the next course of action for the robot surrounded by obstacles.

$$t_c = \frac{dist_o^i(t)}{\mathbf{v}_{ro}}. \quad (6)$$

The position of the obstacle ($\mathbf{P}_{OB}$) with respect to robot position at time t can be determined using:

$$\begin{aligned} \mathbf{P}_{OB} &= [x_o \ y_o]^T \\ &= [x_r \ y_r]^T + dist_o^i(t) \times [\cos \alpha \ \sin \alpha]^T, \end{aligned} \quad (7)$$

where (x_o, y_o) , (x_r, y_r) and α are the current position of the obstacle, current position of the robot and angle made by the relative velocity with respect to robot's velocity at time t , respectively.

To generate a collision free navigation path, the mobile robots need to identify the obstacle position as

well as its teammates (other robots) positions and maintain a safe distance from obstacles. The position of obstacles with respect to the mobile robot can be evaluated using the Eq. (7). Thus, a linear function for path safety can be formed as:

$$f_{\text{obs}}^l(P_r^i) = \begin{cases} 0, & \|P_r^i - P_{OB}^l\| > r_1 \\ \frac{1}{\|P_r^i - P_{OB}^l\|} - \frac{1}{r_1}, & \|P_r^i - P_{OB}^l\| \leq r_1 \end{cases}, \quad (8)$$

where P_r^i and P_{OB}^l are the position of i th mobile robot and l th obstacle for $l = \{1, 2, \dots, m\}$, respectively. To evade the collision with obstacles, the robot should maintain a minimum distance greater than r_1 . Consequently, f_{obs}^l will attain a nonzero value if the distance of obstacle with respect to robot is less than r_1 otherwise zero. During navigation of a mobile robot, it may face more than one obstacle in its surrounding. Hence, the path safety factor for robot relative to obstacles and its teammates can be expressed as:

$$f_{\text{obs}}(P_r^i) = \max_{1 \leq l \leq m} \{f_{\text{obs}}^l(P_r^i)\}. \quad (9)$$

Finally, the objective function for obstacles avoidance can be formed as:

$$f_{\text{obs_avoid}}^i = \min \{f_{\text{obs}}(P_r^i)\}. \quad (10)$$

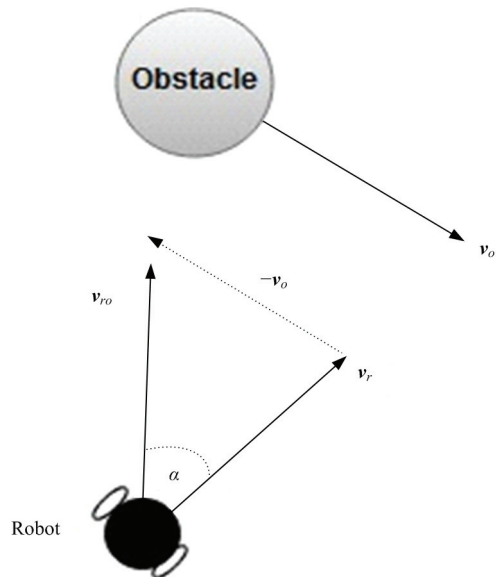


Fig. 5 Relative velocity illustration for the mobile robot and moving obstacle.

Therefore, a single objective fitness function comprising the weighted sum of above discussed objectives can be formulated for multi-robot navigation as:

$$F_i = \gamma_1 f_{\text{path}}^i + \gamma_2 f_{\text{smooth_wp}}^i + \gamma_3 f_{\text{smooth_ang}}^i + \gamma_4 f_{\text{obs_avoid}}^i, \quad (11)$$

where $\gamma_1 = 0.6 \text{ unit} \cdot \text{pixel}^{-1}$, $\gamma_2 = 0.05 \text{ units}$, $\gamma_3 = 0.1 \text{ units} \cdot \text{radian}^{-1}$ and $\gamma_4 = 0.25 \text{ unit pixel}$ are the nonzero positive weight factors for path cost, number of waypoints, angle of deviation and path safety, respectively. The weight factors γ_1 , γ_2 , γ_3 , and γ_4 are the governing parameters of the fitness function for faster convergence. The values of the weight factors are chosen here by trial and error approach.

3 Proposed Hybrid IWD-DE algorithm

3.1 Basic IWD algorithm

Intelligent Water Drops (IWD)^[46] algorithm is a population based meta-heuristic optimization algorithm that mimics the behaviour of water drops which runs in rivers. IWD has the ability to provide impressive solutions to various complex optimization problems. The water drops of a river flow intelligently under the action of earth's gravitational force towards the destination by avoiding different types of obstacles on their path. The absence of obstacles towards the destination makes the water drops to travel along a straight path, which is the Euclidian distance from the source to destination. However, in reality, the water drops in rivers meet several barriers such as mountains, highlands and swamps. The presence of these obstacles influences the water drops to take turns intelligently to reach at the destination by avoiding the obstacles. It has been observed that the path constructed by the water drops in the presence of the obstacles seems to be the shortest path with respect to the distance from source to destination.

The flow of water drops in its environment is characterized by two important properties: (1) the velocity of IWD and (2) the soil amount that the IWD holds. Several numbers of possible path may exist between source and destination in an environment. Now it is important to identify the most suitable one among several choices. The selection of most suitable path by IWD depends on the path hardness. The path hardness is characterized by the amount of soil on the path. The path containing the least amount of soil is more desirable as

compared to those with more amount of soils. The relationship between the velocity of IWD and the amount of soil it holds can be established by considering discrete finite steps taken by IWD from the source to the destination. The velocity of IWD holds an inverse relation with the amount of soil present between two adjacent positions. There exists an inverse relation between the amount of soil carried by the IWD and the time required for the water drops to cover the next positions from its present positions. The mobile robots in the navigation planning problem may meet several obstacles on their way towards the destination. The above discussion reveals that there is a similarity of path finding behaviour between water drops in rivers and navigation planning of mobile robots. This fact motivated us to employ the concept of IWD in the field of mobile robot navigation.

3.2 Mobile robot navigation using IWD

The movement of mobile robot from source to destination may be represented as a graph $G = \{V, E\}$, where V is the set of waypoints treated as vertices in between the source to destination and E represents the sub path connecting to any two waypoints. The waypoints are represented in terms of two dimensional coordinate systems. Each sub path contains some amount of soil. Here, the mobile robot will be treated as an IWD. The IWD now decides the next waypoint based on the concentration of soil in the path. Soil concentration in a path depends on the presence of obstacles along its path. The concentration of soil is indirectly proportional to the distance of IWD from the obstacle which subsequently controls the velocity of robots. IWD stores all the visited waypoints in its path. The proposed IWD algorithm for multi-robot navigation can be formulated as follows.

(a) The initial configuration of environment consisting of n number of robots with known initial and goal positions in a two dimensional coordinates system. Several numbers of obstacles are placed at different positions in the environment. The algorithm is initialized with N^{IWD} number of IWD moving with an initial velocity V_{IWD} . IWD updates its velocity as:

$$V_{\text{IWD}}(t+1) = V_{\text{IWD}}(t) + \frac{\alpha_v}{\beta_v + \eta_v \cdot \text{soil}^2(i, j)}, \quad (12)$$

where $\alpha_v = 1$, $\beta_v = 0.01$ and $\eta_v = 1$ are the constant pa-

rameters for updating the velocity of IWD.

(b) During the journey of IWD it accumulates soil from the sub path. The initial amount of soil in the sub path is $soil_{init}(i, j)$. The amount of soil taken by IWD from the sub path can be expressed as:

$$\delta soil(i, j) = \frac{\alpha_s}{\beta_s + \eta_s \cdot T(i, j; V_{IWD})}, \quad (13)$$

$$T(i, j; V_{IWD}) = \frac{dist(p(i), p(j))}{\max(\varepsilon_v, V_{IWD})}, \quad (14)$$

where $\alpha_s = 1, \beta_s = 0.01$ and $\eta_s = 1$ are the constant parameters for updating the soil amount accumulated by the IWD, $T(i, j; V_{IWD})$ denotes the time taken by the IWD to move from i th waypoints to j th waypoint with the velocity of V_{IWD} , $p(i)$ and $p(j)$ denotes the coordinate point of i th and j th waypoint respectively and $dist()$ denotes the Euclidean distance between $p(i)$ and $p(j)$. The small positive number $\varepsilon_v = 0.001$ is used to prevent the negative value of the velocity.

(c) Further, the amount of soil contained in the sub path and accumulated by the IWD need to be updated. Thus, the updated soil amount in the sub path can and IWD can be expressed as:

$$soil(i, j) = (1 - \mu) \cdot soil(i, j) - \mu \cdot \delta soil(i, j), \quad (15)$$

$$soil_{IWD} = soil_{IWD} + \delta soil(i, j), \quad (16)$$

where μ is a small positive constant selected from (0, 1) and $soil_{IWD}$ is the amount of soil accumulated in IWD.

(d) The primary job of each IWD is to select the next waypoint from its current location. The next waypoint will be selected based on the soil concentration between the current and next waypoint. Less concentration of soil in the sub path between two waypoints preferred the most. Let i be the current waypoint of IWD, then the probability of selecting j th waypoint while moving from i to j is $prob_{i,j}^{IWD}$. Thus, the probability of selecting the next waypoint for IWD can be expressed as:

$$prob_{i,j}^{IWD} = \begin{cases} \frac{f(soil(i, j))}{\sum_{1 \leq k \leq N^{IWD}, j \neq k, k \notin P_{OB}} f(soil(i, k))} & \text{if } j \notin P_{OB} \\ 0 & \text{else} \end{cases}, \quad (17)$$

Such that

$$f(soil(i, j)) = \frac{1}{\varepsilon_s + g(soil(i, j))}, \quad \text{and} \quad (18)$$

$$g(soil(i, j)) = \begin{cases} soil(i, j) & \min_{l \in IWD_{vw}} soil(i, l) \leq 0 \\ soil(i, j) - \min_{l \in IWD_{vw}} soil(i, l) & \text{else} \end{cases}, \quad (19)$$

where P_{OB} is the position of obstacle, ε_s is the small positive constant to prevent division by zero constraint in the function $f()$, the function $g(soil(i, j))$ is used to maintain a positive value to $soil(i, j)$ and IWD_{vw} is the set all visited waypoints in the path.

(e) Each IWD travels along the sub path by deciding their respective next waypoints to construct an optimal and safe path until reaching at the goal position. At the end of each iteration the objective function value is evaluated by Eq. (11) to measure the fitness of each IWD. Then, the iteration best solution is found based on the minimum value of the objective function and can be represented as:

$$T_{ibest} = \min(F(T^{IWD})), \quad (20)$$

where $F()$ denotes the fitness function, the fitness value T^{IWD} evaluated by each IWD given by $F(T^{IWD})$ and T_{ibest} denotes the iteration best solution. Based on quality of T_{ibest} , the corresponding sub paths of the solution will be updated using:

$$soil(i, j) = \rho_s \cdot soil(i, j) + \rho_{IWD} \cdot k(N_c) \cdot soil_{IWD}^{ibest} \quad \forall (i, j) \in T_{ibest}, \quad (21)$$

where $soil_{IWD}^{ibest}$ is concentration of soil by the iteration best IWD, N_c is the number of visited node by the iteration best IWD and $k(N_c) = 1/N_c - 1$. The parameters ρ_s and ρ_{IWD} are constants with positive and negative value respectively. The term $soil(i, j)$ in the right side of Eq. (21) represents the soil amount remains from previous iteration. The value of global best solution (T_{gbest}) is updated after each iteration by:

$$T_{gbest} = \begin{cases} T_{gbest} & \text{if } F(T_{gbest}) \geq F(T_{ibest}) \\ T_{ibest} & \text{otherwise} \end{cases}. \quad (22)$$

3.3 Hybridization of IWD-DE approach for multi-robot navigation planning

Differential Evolution (DE)^[47] is a novel heuristic approach for global optimization, which is used to

minimize the nonlinear and non-differentiable continuous space functions. DE algorithm uses three basic operators, mutation, crossover and selection for global optimization. Initially, DE generates an initial solution randomly in the search space and adds the difference of two population vector to the initial solution to generate a trial vector. Then, DE algorithm combines the result of trial vector with the target vector to generate a new individual. The fitness value of new individual and the target vector is compared and the vector with better fitness is replaced by the other for the next generation. The basic idea of hybrid IWD-DE algorithm is to combine the positive attribute of both the algorithms to find a better solution. DE algorithm has the ability to provide better solution among many with the help of selection mechanism.

The primary concern in navigation planning is to navigate the robot in a shortest path such that it will reach its target quickly and safely. Thus, IWD algorithm is employed in multi-robot navigation problem to achieve the above objective in a dynamic environment. However, the observation made from many runs confirms the slower convergence of the IWD algorithm and does not guarantee an optimal solution always in the presence of dynamic obstacle. The soil concentration on the path plays an important role in maintaining the exploration and exploitation of the path. A rational distribution of the soil amount in the path can directly affect IWD to explore their optimal paths. Thus, the operators of DE algorithm are used to make some random deviation disturbance in the soil concentration of the path. The intention of incorporating DE is to make a reasonable disturbance of soil concentration in the path, which leads the IWD to find an optimal path. In the proposed IWD-DE algorithm, the results obtained from Eqs. (21) and (22) of IWD are given as an input to DE. The mutant vectors are generated in the mutation operation of DE as:

$$M_{\text{soil}}^k(i, j) = T_{\text{gbest}} + F \cdot (\text{soil}_{q_1}(i, j) - \text{soil}_{q_2}(i, j)), \quad (23)$$

$$k = 1, 2, \dots, N^{\text{IWD}} \text{ and } q_1 \neq q_2 \neq k,$$

$$F = F_{\text{max}} - (F_{\text{max}} - F_{\text{min}}) \cdot \left(\frac{\text{iter}_{\text{max}} - \text{iter}}{\text{iter}_{\text{max}}} \right), \quad (24)$$

where T_{gbest} the best value of soil concentration obtained from IWD, q_1 and q_2 are randomly chosen mutual ex-

clusive integers from $[1, N^{\text{IWD}}]$. F is an amplification factor. The parameter F is a real number called mutant factor and it controls the amplification of difference vector, F_{min} and F_{max} are the minimum and maximum values of the amplification factor which ranges from $[0.5, 1]$. Crossover operation is then employed to determine the trial vector as:

$$T_{\text{soil}}(i, j) = \begin{cases} M_{\text{soil}}(i, j), & \text{rand}_j \leq CR \\ \text{soil}(i, j) & \text{otherwise} \end{cases}, \quad (25)$$

$$CR = CR_{\text{max}} - (CR_{\text{max}} - CR_{\text{min}}) \left(1 - \frac{\text{iter}}{\text{iter}_{\text{max}}} \right). \quad (26)$$

The selection mechanism is finally employed to provide the best solution vector for the next iteration as:

$$\text{soil}(i, j) = \begin{cases} T_{\text{soil}}(i, j), & F(T_{\text{soil}}(i, j)) < F(\text{soil}(i, j)) \\ \text{soil}(i, j), & \text{otherwise} \end{cases}. \quad (27)$$

The approach for multi-robot navigation by employing Hybrid IWD-DE algorithm in a dynamic environment is presented in Algorithm 1 (see Appendix). The proposed algorithm is implemented using a distributed approach. The source and goal position of each robot in the environment is given as input to Algorithm 1 and it calls the procedure $\text{IWD-DE}(x_{\text{curr}-i}, y_{\text{curr}-i})$ to decide the succeeding waypoint of each robot until all the robot reached their destination. The pseudo code of the procedure $\text{IWD-DE}(x_{\text{curr}-i}, y_{\text{curr}-i})$ is defined in Algorithm 2 (see Appendix). The procedure $\text{IWD-DE}(x_{\text{curr}-i}, y_{\text{curr}-i})$ evaluates the objective function defined in Eq. (11) to determine the successive waypoint for every robot.

4 Performance evaluation and discussion

Preserving the ultimate objective, the efficiency of the suggested algorithm for multi-robot navigation has been verified through simulation and experimentation in real platform with Fir bird robot in a dynamic workspace. The outcomes of the experimentation are discussed below.

4.1 Simulation analysis

The simulation environment for multi-robot navigation problem is configured through C language. The simulation is carried out on a desktop PC with Intel Core

i5 processor, 4 GB of RAM and 500 GB HDD. Every mobile robot in simulation is denoted by a circle with radius of 6 pixels. The start and target position for each robot is pictured with circle and square shape, respectively. In the present simulation, the environment is consisting of 8 mobile robots and 11 polygonal obstacles out of which 8 obstacles are dynamic. During the execution of simulation, the maximum velocity of every mobile robots is set to $12 \text{ pixels} \cdot \text{s}^{-1}$. The current position of individual mobile robot is taken as an input to the algorithm. The algorithm outputs the subsequent way-point for individual robot by evaluating the fitness

function stated in Eq. (11) until the robots reached their respective targets. The simulation is executed over 30 runs with 100 iterations per run and the performance of the algorithm is verified through average of all runs.

The initial setup of the workspace for simulation is shown in Fig. 6a. The final configuration of robots by employing IWD-DE, IWD and DE is presented in Figs. 6b – 6d, respectively. It can be observed that the IWD-DE, IWD, and DE algorithms take 38, 45, and 51 numbers of iteration respectively, for all the robots to arrive at their respective destinations. Thus, the performance of IWD-DE algorithm in simulation is

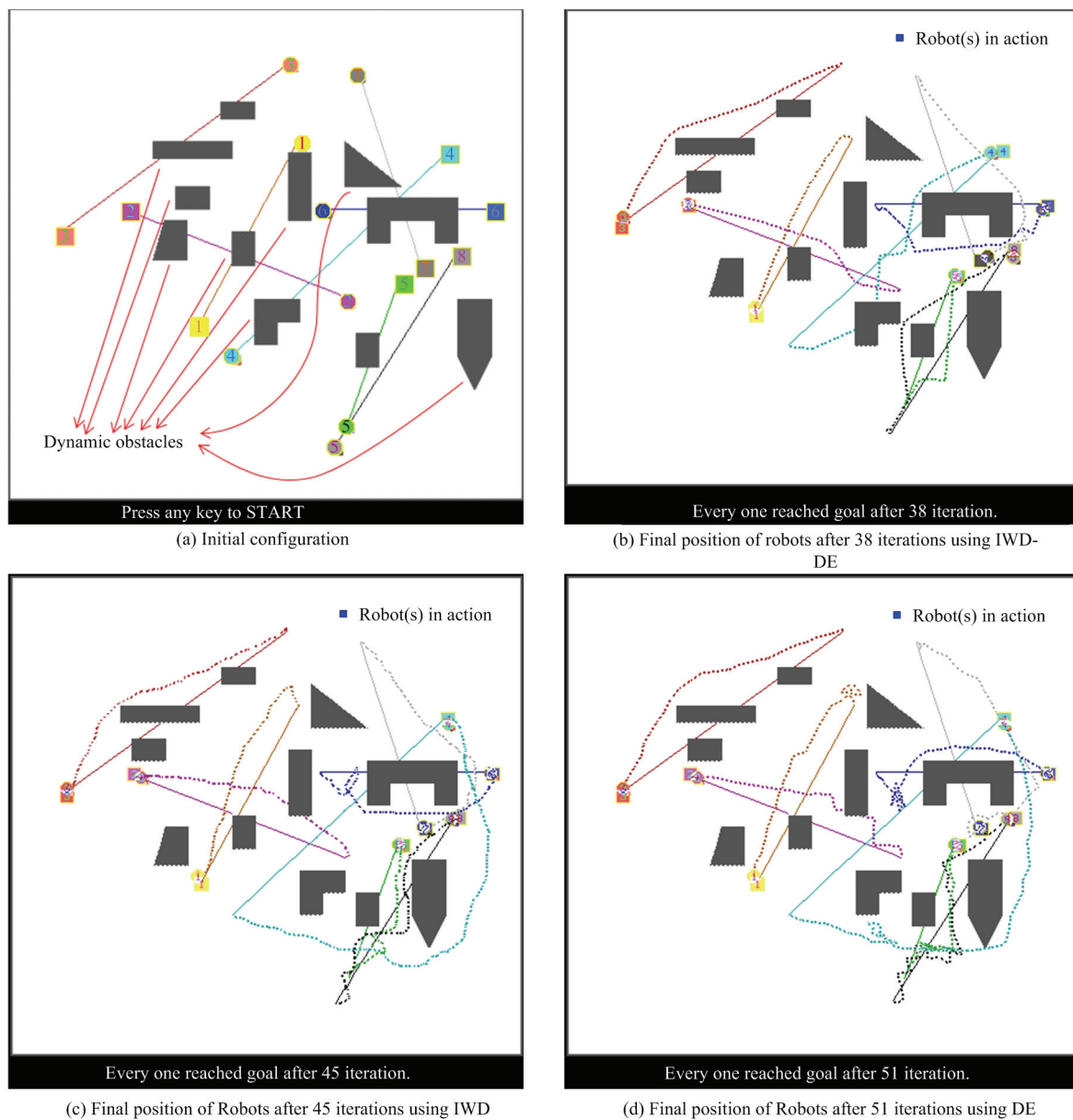


Fig. 6 Navigation planning of mobile robots by employing IWD-DE, IWD and DE with 8 mobile robots and 11 obstacles.

better than those of basic IWD and DE algorithms in terms of number of iterations. Further, the individual robot performance in terms of number of iterations by employing IWD-DE, IWD and DE algorithms is presented in Table 1. It can be noted from Table 1 that all the robots except Robot-2 arrived their respective target positions in less number of iterations using IWD-DE algorithm as compared to the basic IWD and DE algorithms. However, Robot-2 takes a less number of iterations using the basic IWD algorithm as compared to IWD-DE and DE algorithms.

The performance of algorithms is further investigated using some additional parameters such as “Aggregate Navigation Path Deviation (ANPD)”, “Average Navigation Path traveled (ANPT)”, “Average Unexplored Goal Distance (AUGD)”, number of turns and execution time. The definition of these parameters can be found in Ref. [48]. The variation of ANPT and ANPD with number of robots for different algorithms are depicted in Figs 7 and 8, respectively. The perceptions made through Figs. 7 and 8 reveal the preeminence of IWD-DE algorithm as compared to the basic IWD and DE algorithms.

The individual robot performance in terms of AUGD by employing IWD-DE, IWD and DE algorithms is illustrated in Figs. 9 – 11, respectively. Further, the overall performance of all the mobile robots by employing the algorithms in the presence of 8 dynamic obstacles and 3 static obstacles in terms of AUGD is presented in Fig. 12. It tends to be noticed that the IWD-DE algorithm attains a zero value with less number of iterations as those of basic IWD and DE algorithms.

The robustness of the IWD-DE algorithm is further

Table 1 Number of iterations taken by individual robots to arrive at their targets by different algorithms in a dynamic environment

Robot number	Number of iterations to reach target		
	IWD-DE	IWD	DE
R_1	24	28	31
R_2	27	26	29
R_3	33	34	35
R_4	38	45	51
R_5	19	23	28
R_6	24	36	44
R_7	29	30	36
R_8	28	32	38

quantified through different performance metrics such as iterations, execution time, ANPT and ANPD. A wheeled mobile robot utilizes its energy due to the onboard chips, circuits, sensory equipment and kinetic energy^[49]. However, it mostly spends its energy due to motion as compared to other sources of energy and thus it is ignored here. The utilization of energy of a mobile robot is more in angular motion as compared to linear motion. Hence, the number of turns is considered as one of the additional parameters for performance evaluation. The performance of algorithms in terms of the above discussed parameters by varying the number of robots and obstacles is noted and presented in Table 2. The outcome proves the superiority of the IWD-DE

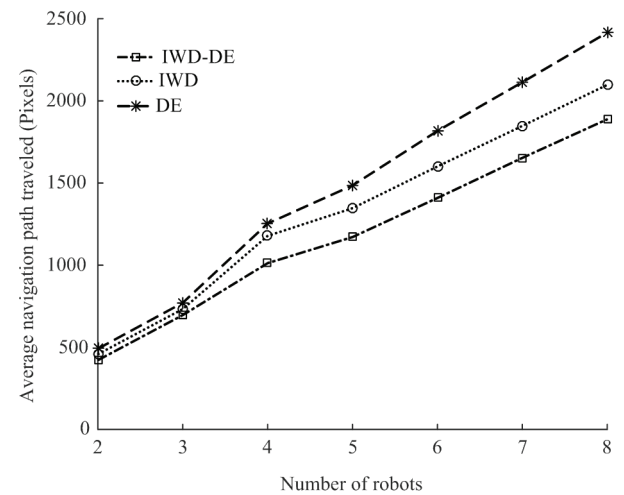


Fig. 7 ANPT versus number of robots in the presence of 8 dynamic and 3 static obstacles.

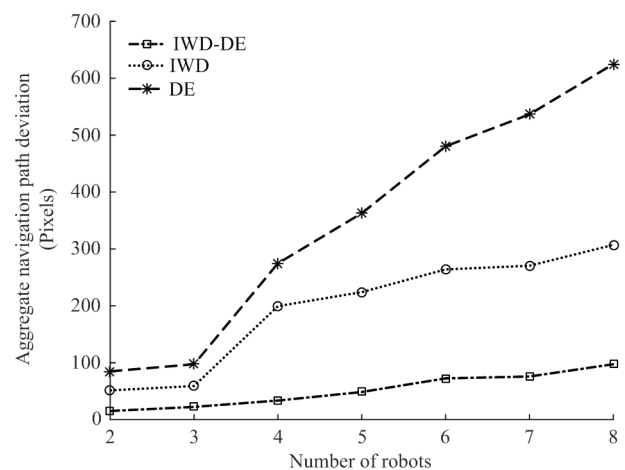


Fig. 8 ANPD versus number of robots in the presence of 8 dynamic and 3 static obstacles.

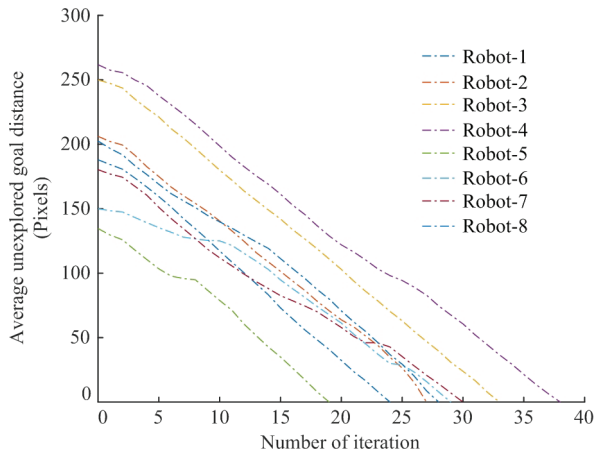


Fig. 9 AUGD versus number of iterations of the individual mobile robot using IWD-DE algorithm in the presence of 8 dynamic and 3 static obstacles.

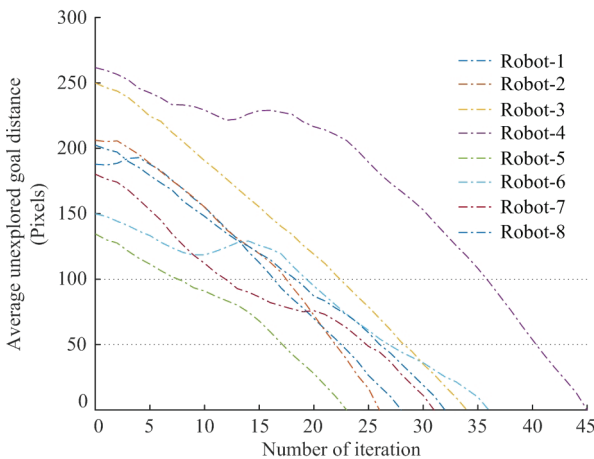


Fig. 10 AUGD versus number of iterations of the individual mobile robot using IWD algorithm in the presence of 8 dynamic and 3 static obstacles.

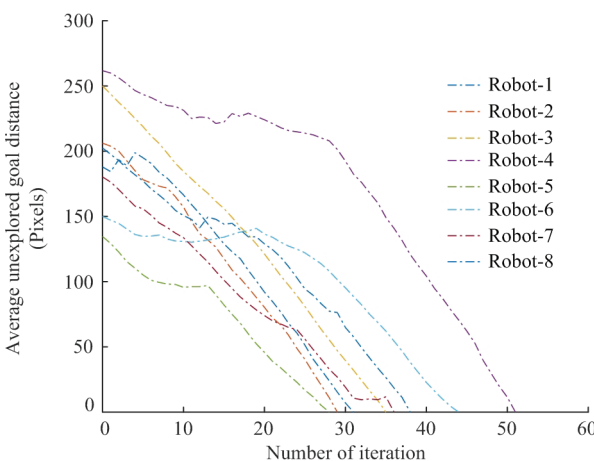


Fig. 11 AUGD versus number of iterations of the individual mobile robot using DE algorithm in the presence of 8 dynamic and 3 static obstacles.

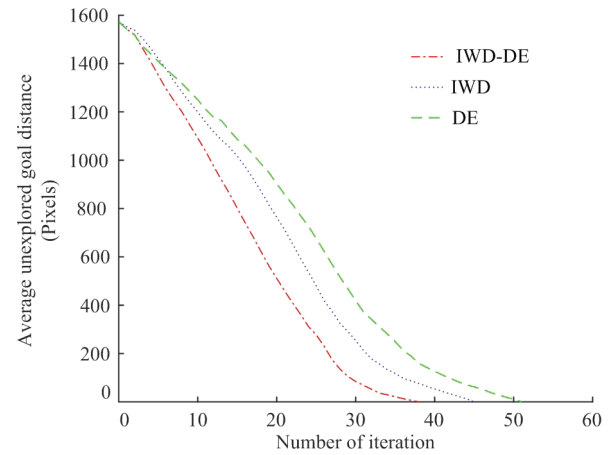


Fig. 12 AUGD versus number of iterations of 8 mobile robots using different algorithms in the presence of 8 dynamic and 3 static obstacles.

algorithm in contrast to basic IWD and DE algorithm in the dynamic workspace. Furthermore, it can be noted, IWD-DE outperforms basic IWD and DE algorithm as the dynamic obstacle increases in the workspace.

4.2 Experimental analysis

The effectiveness of proposed approach is further experimented in a real platform by employing the Fire Bird V robot. The detailed specification of Fire Bird V robot can be found in Ref. [50]. To deal with the uncertainties in perceiving data from the environment, the Fire bird robot is equipped with 8 IR proximity sensors. To maintain the accuracy in perceiving data from environment, the IR sensors are calibrated before starting the experimentation. The Fire Bird robot is interfaced with PC through wireless module where, PC acts as master and robot acts as slave.

The workspace for performing the experimentation is shown in Fig. 13a. The workspace consists of two mobile robots (Fire Bird V) and 9 obstacles out of which 8 are static obstacle and one is dynamic obstacle. The target position of robots in the workspace is shown by two different colored flags. The experimentation was then started by employing IWD-DE algorithm and intermediate position of robot in the workspace is shown in Fig. 13b. The final configuration of workspace by employing IWD-DE, IWD and DE algorithm have been presented in Fig. 13c. The path traveled by individual robot to arrive at their destination by employing the algorithms is presented in different color code. A similar

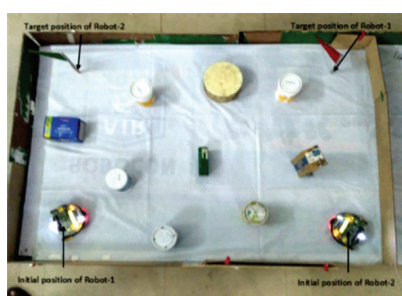
Table 2 Comparison of IWD-DE, IWD and DE by varying the number of robots and obstacles

(a) Comparison of algorithms in terms of number of iteration, execution time and number of turns

No. of robot	No. of obstacles = 12		No. of iteration			Execution time (s)			Number of turns		
	Static obstacles	Dynamic obstacles	IWD-DE	IWD	DE	IWD-DE	IWD	DE	IWD-DE	IWD	DE
2	12	0	21	20	25	22.43	21.84	28.18	8	8	10
	6	6	26	28	30	22.19	24.14	31.72	7	10	13
	0	12	28	30	33	24.4	30.54	34.65	6	13	16
4	12	0	31	31	36	26.6	26.11	30.36	12	24	27
	6	6	36	39	44	30.48	32.87	35.27	17	28	35
	0	12	38	45	51	34.12	38.86	42.23	18	34	41
6	12	0	31	32	37	31.3	31.68	35.11	21	46	44
	6	6	34	41	46	32.21	36.26	40.66	25	49	54
	0	12	39	47	54	35.69	40.37	44.11	27	55	61
8	12	0	33	34	38	32.11	33.13	37.24	31	61	74
	6	6	36	41	48	33.23	37.28	40.61	34	67	81
	0	12	41	49	56	37.47	41.24	44.86	36	80	93

(b) Comparison of algorithms in terms of ANPT and ANPD

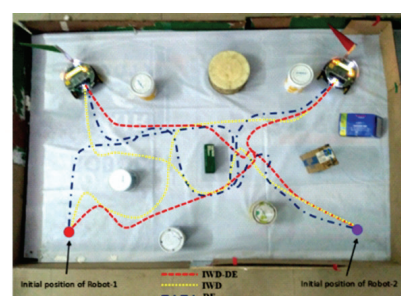
No. of robot	No. of obstacles = 12		ANPT			ANPD		
	Static obstacles	Dynamic obstacles	IWD-DE	IWD	DE	IWD-DE	IWD	DE
2	12	0	415.51	411.03	444.61	8.91	4.43	38.01
	6	6	420.86	444.95	466.81	14.26	38.35	60.21
	0	12	445.17	479.43	502.5	38.57	72.83	95.90
4	12	0	996.22	1008.05	1105.42	17.26	29.09	126.46
	6	6	1010.97	1150.44	1205.65	32.01	171.48	226.69
	0	12	1034.71	1266.91	1305.74	55.75	287.95	326.78
6	12	0	1369.27	1508.56	1709.51	32.48	171.77	372.72
	6	6	1406.54	1583.85	1811.87	69.75	247.06	475.08
	0	12	1478.65	1672.73	1901.55	141.86	335.94	564.76
8	12	0	1813.11	2028.31	2207.35	22.43	237.63	416.67
	6	6	1875.15	2079.98	2382.52	84.47	289.3	591.84
	0	12	1911.77	2204.37	2596.95	121.09	413.69	806.27



(a) Initial configuration



(b) Intermediate configuration



(c) Final configuration

Fig. 13 Navigation planning of two mobile robots in real environment.

environment as compared to the real environment is created using simulation and presented in Fig. 14a. The performance of the algorithms in the simulation environment is studied and presented in Figs. 14b – 14d.

Fig. 14b represents the final configuration of mobile robots where both mobile robots reached their predefined target position after 59 iterations by employing IWD-DE algorithm. Figs. 14c and 14d represent the

final configuration of mobile robots after 71 and 76 iterations by employing IWD and DE, respectively. Finally, the comparison has been carried out for simulation as well as experimental results for the same environment and its result has been presented in Table 3. The conclusion drawn from Table 3 shows that IWD-DE generates the results in terms of ANPT, ANPD and number of turns with greater accuracy compared to IWD and DE in both environments.

5 Comparative analysis with state-of-the-art

The comparison of proposed IWD-DE algorithm with the state-of-the-art^[27,51] is discussed in this section. Firstly, the potential of the IWD-DE is verified against the FLC^[27] through simulation outcomes, conducted in a similar environment. The workspace consists of three mobile robots and 6 obstacles, out of which 4 obstacles are dynamic and its direction of motion is shown in Fig. 15a with different color codes. The maximum velocity of robots and dynamic obstacles are set to $20 \text{ pixels}\cdot\text{s}^{-1}$ and $10 \text{ pixels}\cdot\text{s}^{-1}$, respectively. The best

simulation results for 30 runs by employing IWD-DE and FLC approach is shown in Figs. 15b and 15c respectively. The performance of the algorithms is quantified in terms of ANPT, ANPD and execution time on the average of 30 runs. The value of ANPT, ANPD and execution time for the proposed approach is reaching 1391.06 pixels, 109.17 pixels and 39.42, while, it is reaching 1525.48 pixels, 243.59 pixels and 46.87 for FLC^[27]. Thus, the IWD-DE approach achieves an improvement of 8.81%, 55.18% and 15.89% over FLC in terms of ANPT, ANPD and execution time, respectively.

The learning based algorithms recently gain much attention in the domain of mobile robot navigation. Thus, the second part includes the state-of-the-art^[51] for a fair comparison with the proposed IWD-DE approach. In order to implement Multi Agent Reinforcement Learning (MARL), the workspace is considered as a virtual grid map and the learning phase of robot is initiated through various test cases. In every test scenario the workspace is added with new obstacles and the learning process is continued until the robots reached their target,

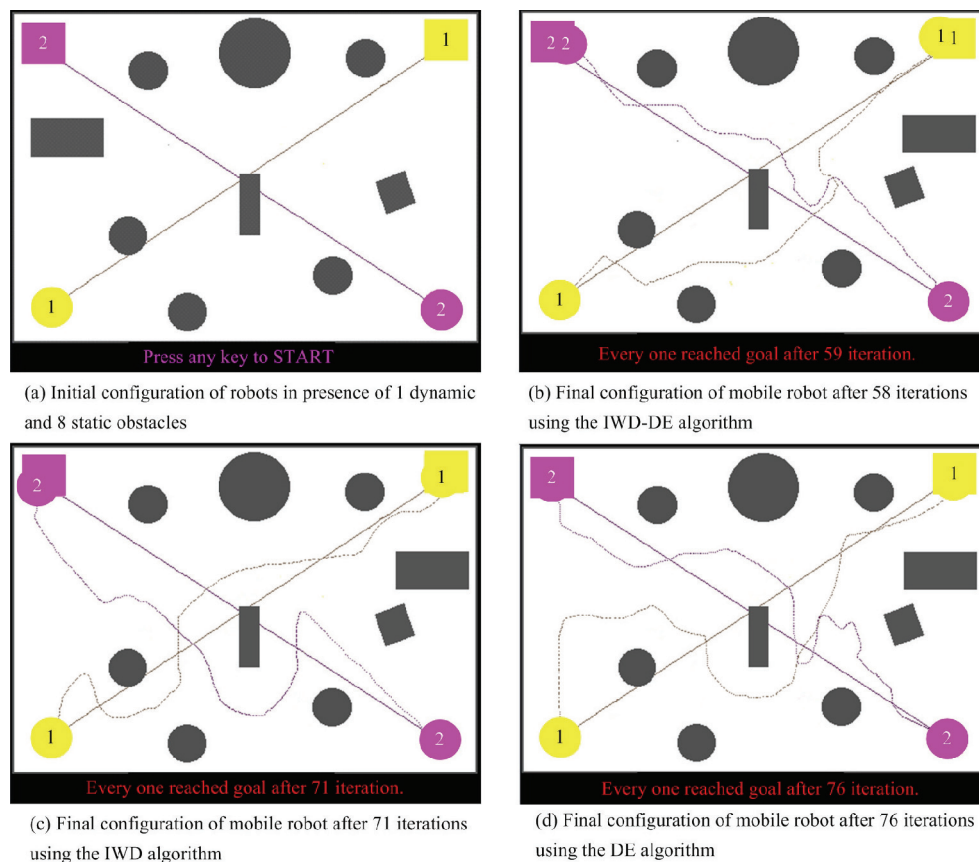


Fig. 14 Multi-robot navigation planning in simulation environment equivalent to real environment.

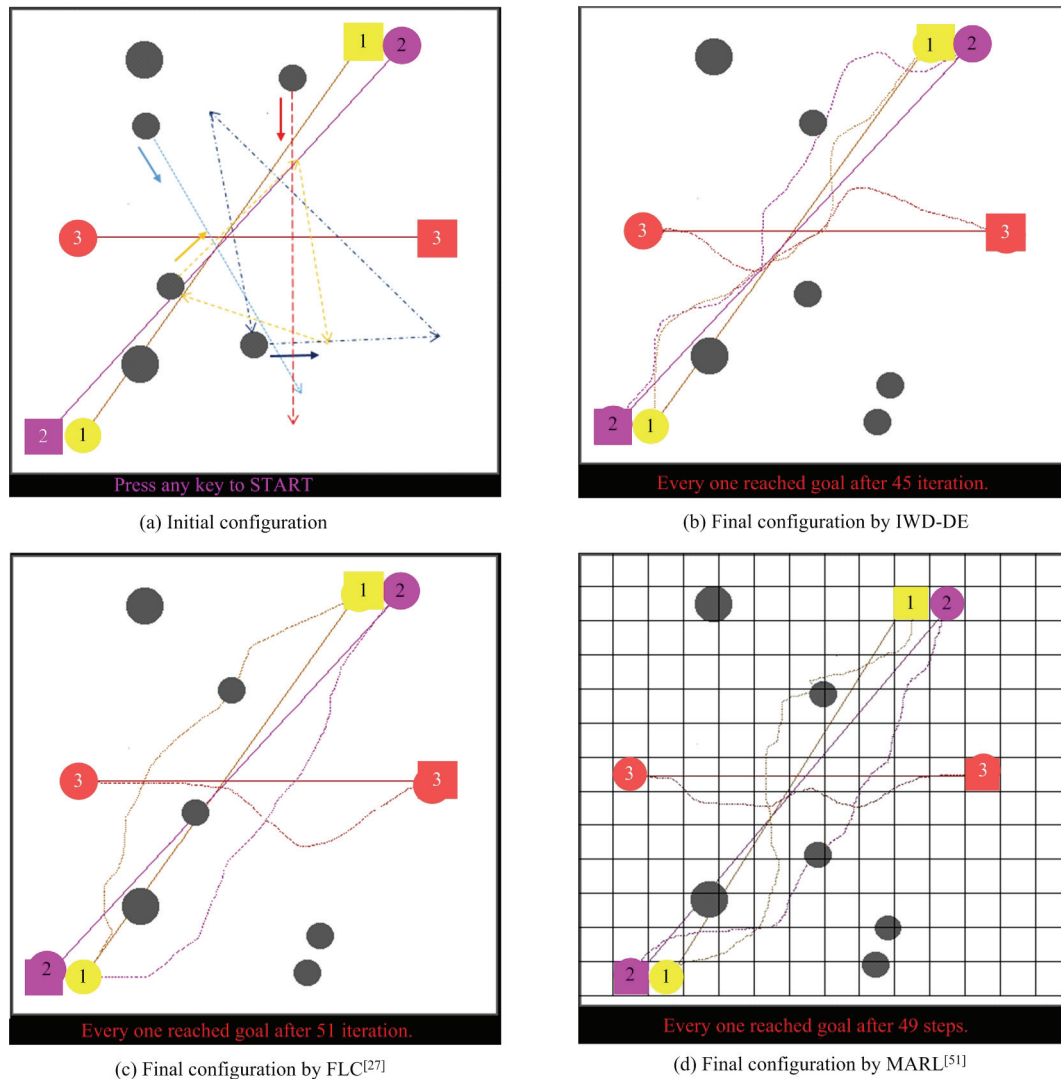


Fig. 15 Multi-robot navigation planning in simulation environment by employing IWD-DE, FLC^[27], and MARL^[51].

Table 3 Comparison of simulation and experimental result of two mobile robots with 9 obstacles comprise of one static and 8 dynamic obstacles

Algorithm	Simulation result (Pixels)			Experimental result (Pixels)			% of error		
	ANPT	ANPD	No. of turns	ANPT	ANPD	No. of turns	ANPT	ANPD	No. of turns
IWD-DE	1102.75	145.18	28	1131.30	173.73	30	8.96	16.43	6.66
IWD	1275.95	318.38	44	1403.57	446.00	49	9.09	28.61	10.20
DE	1376.67	419.10	48	1549.17	591.60	56	11.13	29.15	14.28

or maximum iteration reached, or they collide with any obstacles or teammates. Accordingly, the Q-table is updated and the above step is continued until the convergence of Q-table. The kernel training is then started using the values of Q-table to anticipate an approximate action by the agent when it finds a new state in the environment. The simulation is carried out for 30 runs and the best performance by employing MARL approach is

shown in Fig. 15d. In the simulation study, it has been observed that the IWD-DE and FLC approach are succeeding in all runs. However, MARL succeeds on 21 runs out of 30 runs. MARL approach spends lots of time in learning and also incompetent as the number of dynamic obstacles increase in the workspace. Every agent in MARL approach needs to refer the Q-table to determine the best possible action from the set of actions in

the current state. Thus, the computational cost of MARL includes the cost for accessing Q-table and selection of best possible action. However, the proposed IWD-DE approach can be employed to any unknown environment and cost of computation is also less as compared to MARL.

The simulation outcomes of MARL in terms of ANPT, ANPD and execution time over the average of 30 runs are 1466.37 pixels, 184.48 pixels, and 44.19, respectively. The results confirm the improvement of proposed IWD-DE over MARL by 5.13%, 40.82% and 10.79% in terms of ANPT, ANPD and execution time, respectively. Therefore, the proposed IWD-DE approach is more robust and competent as compared to the state-of-the-art in Refs. [27] and [51].

6 Conclusion and future scope

In the current study, a novel hybrid IWD-DE algorithm for multi-robot navigation in dynamic workspace has been proposed. A single objective fitness function comprising weighted sum of path cost, smoothness of path and path safety has been formed. A novel hybrid IWD-DE algorithm has been proposed and employed successfully in multi-robot navigation problem. The proposed algorithm is able to find optimal waypoints for individual robot towards their respective target position by evaluating the fitness function iteratively. The potency of the proposed algorithm is verified through simulation as well as with real robot in a dynamic workspace. The efficiency of the proposed algorithm is examined through various influential parameters such as number of iterations, number of turns, execution time, ANPT, ANPD and AUGD. The performance evaluation of the proposed IWD-DE algorithm based on these parameters proves its superiority as those of basic IWD, DE. To have a fair comparison, the potency of the proposed algorithm is further investigated with existing state-of-the-art. The outcome of the comparison also demonstrates the dominance of the proposed approach over the state-of-the-art. However, the effectiveness of the proposed approach in real environment by employing more number of robots has not been verified. Further, improvement on these algorithms to avoid deadlock situation among robots is one of the scope for future study.

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Appendix

Algorithm 1: Algorithm for navigation planning using IWD-DE algorithm

Input: (x_0^i, y_0^i) , (x_{wp+1}^i, y_{wp+1}^i) are the initial and goal position of i th robot respectively where, $1 \leq i \leq n$ and ε is the threshold cost, number of runs (k) and maximum iteration ($iter_max=100$)

Output: safe navigation path NP_i for each robot R_i from (x_0^i, y_0^i) to (x_{wp+1}^i, y_{wp+1}^i)

```

1      : Begin
2      :   For run = 1 to  $k$ 
3      :       For robot  $i = 1$  to  $n$ 
4      :           Update  $x_{curr-i} \leftarrow x_0^i$ ,  $y_{curr-i} \leftarrow y_0^i$ 
5      :       End For
6      :        $iter = 1$ 
7      :       While  $(\|C_i - G_i\| > \varepsilon \ \& \ iter \neq iter\_max)$  // where,  $C_i = (x_{curr-i}, y_{curr-i})$ 
                                and  $G_i = (x_{wp+1}^i, y_{wp+1}^i)$  //
8      :           For robot  $i = 1$  to  $n$ 
9      :               Call  $IWD-DE(x_{curr-i}, y_{curr-i})$ 
10     :           End For
11     :           For  $i = 1$  to  $n$ 
```

```

12      :           Move to  $(x_{\text{next-}i}, y_{\text{next-}i})$ 
13      :           End For
14      :            $iter = iter + 1$ 
15      :       End While
16      :   End for
17      End

```

Algorithm 2: Pseudo code for procedure $IWD-DE(x_{\text{curr-}i}, y_{\text{curr-}i})$

Initialize: Number of IWD ($N^{\text{IWD}} = 50$), initial soil between two nodes ($init_soil = 100$), initial velocity of IWD ($init_vel = 10$), initial fitness value of global best solution ($T_{\text{gbest}} = -\infty$), visited waypoint list $IWD_{vw} = \{ \}$, initial soil in each IWD ($soil_{\text{IWD}}=0$).

```

1      For  $i = 1$  to  $N^{\text{IWD}}$ 
2      :   The visited waypoint list  $IWD_{vw}$  is updated by the current positions of IWD.
3      :   End for
4      :   For  $i = 1$  to  $N^{\text{IWD}}$ 
5      :       Chose the next waypoint  $j$  to be visited by IWD using Eq. (17)
6      :       Update velocity of IWD to move from waypoint  $i$  to  $j$  using Eq. (12)
7      :       Compute the amount of soil taken by IWD with current velocity using Eq. (13)
8      :       Update the soil concentration in the sub path traversed by the IWD using Eq. (15)
9      :   End for
10     :   Update the iteration best solution using Eq. (20)
11     :   Update the soil concentration in the sub path based on iteration best solution using Eq. (21)
12     :   Update the global best solution using Eq. (22)
13     :   Mutation, crossover and selection operation of DE are then carried out using Eqs. (23), (25) and (27) re-
        :   spectively to evaluate the next position of IWD.
14     :   Return  $(x_{\text{next-}i}, y_{\text{next-}i})$ 

```